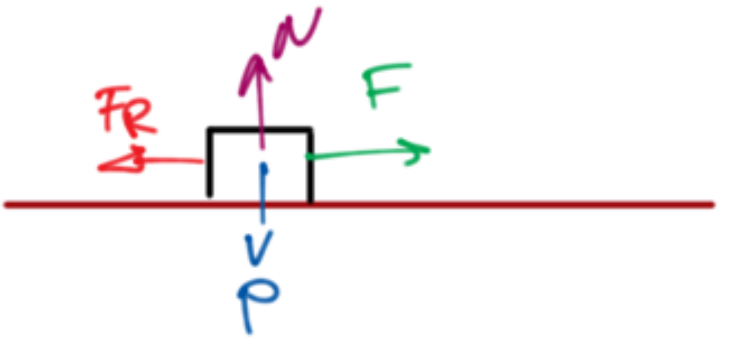
① 

$m = 80\,kg$
$F = 320\,N$
$\mu = ?$
$g = 10\,m/s^2$

Si $v = cte$, $a = 0$

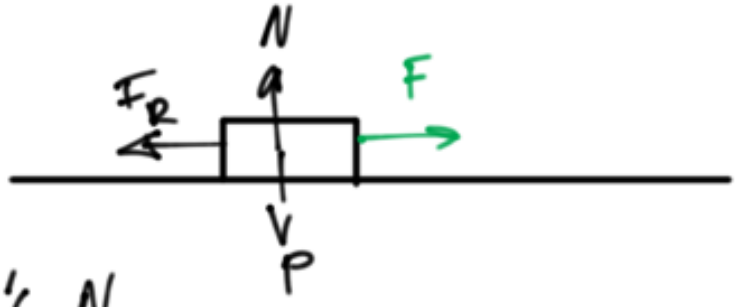
a) b)

$$F - F_R = 0 \longrightarrow F = F_R = 320N$$
$$F - mg\mu = 0$$

$$\mu = \frac{F}{mg} = \frac{320}{80 \cdot 10} = 0'4$$

c) $$F - \mu mg = ma$$

$$F = \mu mg + ma = 0'4 \cdot 80 \cdot 10 + 80 \cdot 0'2$$
$$= 336N$$

② 

$m = 0'2\ kg$

$g = 10\ m/s^2$

$F_e = 0'6\ N$

$F_D = 0'4\ N$

a) Valor máximo $F_R = 0'6 N$

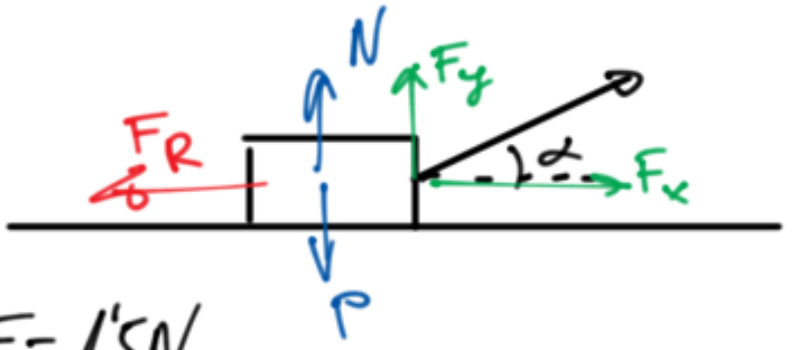
b) $F_R = \mu_e m g = 0'6$

$$\mu_e = \frac{0'6}{0'2 \cdot 10} = 0'3$$

c) $F = F_R = 0'4\ N$

d) $$\mu_D = \frac{0'4}{0'2 \cdot 10} = 0'2$$

(3)



$\alpha = 25°$

$m = 0'3\ kg$

$F = 1'5N$

a) $F_x = F\cos\alpha$

$F_y = F\sin\alpha$

$\sum \vec{F} = m\vec{a}$

Ejey: $N + F_y + P = 0$

$N + F\sin\alpha - mg = 0$

$N = -F\sin\alpha + mg$

$F = 1N$; $N = -1\sin 25 + 0'3 \cdot 10 = 2'58N$

$F = 1'5N$; $N = -1'5\sin 25 + 0'3 \cdot 10 = 2'36N$

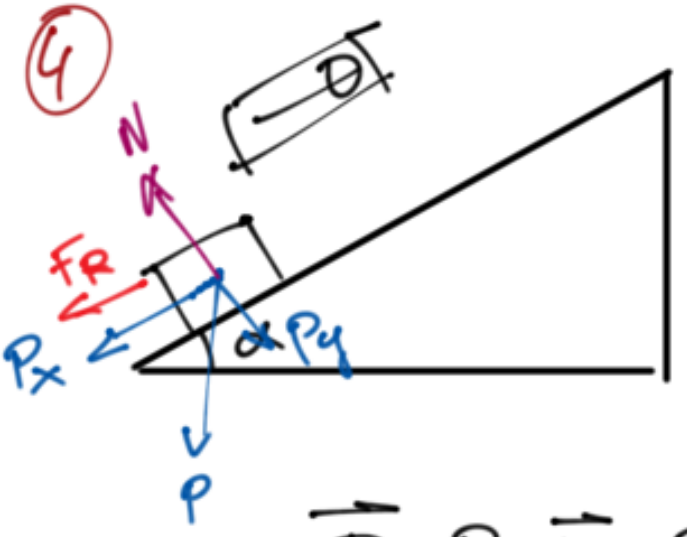
b) $\mu_e = ?$ Se mueve cuando $F = 15$

Eje x: $F_x + F_R = 0$

$$F\cos\alpha - \mu mg = 0$$

$$\mu = \frac{F\cos\alpha}{mg} = \frac{1'5 \cdot \cos 25}{0'3 \cdot 10} = 0'45$$

(4)



$\alpha = 25°$

$v_0 = 5'6\ m/s$

$\mu = 0'4$

$$\vec{P} = P_x\,\vec{i} + P_y\,\vec{j} = -mg\sin\alpha\,\vec{i} - mg\cos\alpha\,\vec{j}$$

$$\vec{N} = -P_y\,\vec{j} = mg\cos\alpha\,\vec{j}$$

$$\vec{F}_R = -\mu mg\cos\alpha\,\vec{i}$$

$$\sum \vec{F} = m\vec{a}$$

Eje x: $-mg\sin\alpha - \mu mg\cos\alpha = ma$

Eje y: $mg\cos\alpha - \mu mg\cos\alpha = 0$

$$a = -g(\sin\alpha + \mu\cos\alpha) = -10(\sin 25 + \cos 25)$$

$$a = -7'85\ m/s$$

$$v(t) = 5'6 - 7'85\,t$$

$$x(t) = 5'6\,t - \frac{7'85}{2}t^2$$

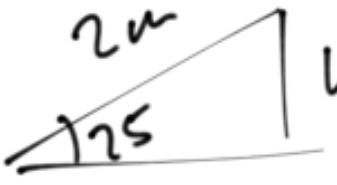
b) Altura a la que se para:

$v = 0$; $\quad 0 = 5'6 - 7'85\,t$

$$t = 0'71 \text{ s}$$

$$x = 5'6 \cdot 0'71 - \frac{7'85}{2}\,0'71^2$$

$$x = 2 \text{ m}$$



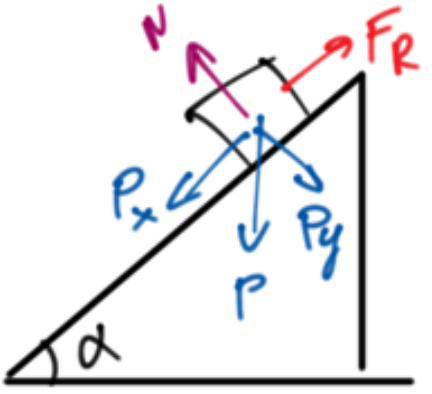
$$\sin 25 = \frac{h}{2}\,; \quad h = 0'84 \text{ m}$$

c) Para que no descienda

ejex: $-mg\sin\alpha + \mu_e mg\cos\alpha = 0$

$\mu_e = tg\alpha = tg\,25 = 0'47$

(5)



$m = 0'4\,kg$

$\alpha = 30°$

$\mu_e = 0'65$

$$\vec{P} = -mg\sin\alpha\,\vec{i} - mg\cos\alpha\,\vec{j}$$

$$\vec{N} = mg\cos\alpha\,\vec{j}$$

$$\vec{F_R} = \mu\, mg\cos\alpha\,\vec{i}$$

$$\sum \vec{F} = m\vec{a}$$

Para que se quede ↓

eje x: $-mg\sin\alpha + \mu mg\cos\alpha = 0$

Si $F_R \geq P_x \rightarrow$ Se queda quieto

Si $F_R < P_x \rightarrow$ Desciende

$P_x = mg\sin\alpha = 0'4 \cdot 10 \cdot \sin 25 = 1'69\,N$

$F_R = 0'65 \cdot 0'4 \cdot 10 \cdot \cos 25 = 2'35\,N$

→ Permanece quieto

b) Si se aplica una fuerza ascendente

$P_x + F_R + F = 0$ F mínima para moverse

$-mg\sin\alpha - \mu mg\cos\alpha + F = 0$

$F = mg(\sin\alpha + \mu\cos\alpha)$

$F = 0'4 \cdot 10(\sin 30 + 0'65\cos 30)$

$F = 4'25\,N$

Si $F \leq 4'25\,N$, no se mueve

Si $F > 4'25\,N$, asciende

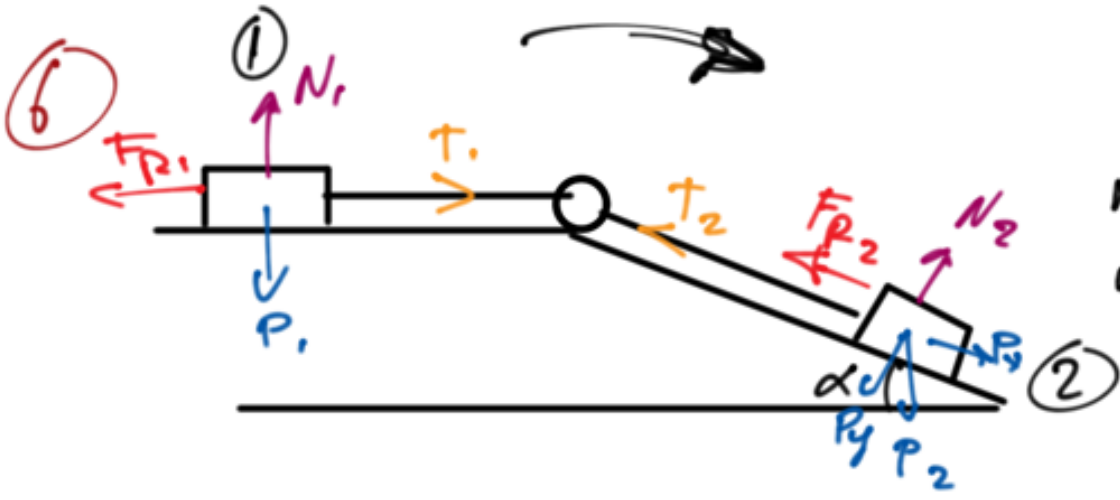
c) $\mu_D = 0'45$

$$-mg\sin\alpha - \mu_D mg\cos\alpha + F = ma$$

$$a = -g(\sin\alpha + \mu_D\cos\alpha) + \frac{F}{m}$$

$$a = -10(\sin 30 + 0'45\cos 30) + \frac{5}{0'4}$$

$$a = 3'6\ m/s^2$$



$\alpha = 30°$

$m_1 = 1\,kg$

$m_2 = 2\,kg$

$\mu_c = 0'35$

$a = ?$

$T = ?$

① $T + F_{R1} = m_1 a$

② $P_x + F_R + T = m_2 a$

① $T - \mu m_1 g = m_1 a$

② $m_2 g \sin\alpha - \mu m_2 g \cos\alpha - T = m_2 a$

$$-\mu m_1 g + m_2 g \sin\alpha - \mu m_2 g \cos\alpha = (m_1 + m_2) a$$

$$-\mu m_1 g + m_2 g (\sin\alpha - \mu \cos\alpha) = (m_1 + m_2) a$$

$$a = g \, \frac{-\mu m_1 + m_2 (\sin\alpha - \mu \cos\alpha)}{m_1 + m_2}$$

$$a = 10 \cdot \frac{-0'35 \cdot 1 + 2(\sin 30 - 0'35 \cos 30)}{1 + 2} =$$

$$= 0'15 \; m/s^2$$

$$T + F_{R1} = m_1 a$$

$$T - m_1 g \mu = m_1 a$$

$$T = m_1 (g\mu + a) = 1\,(10 \cdot 0'35 + 0'15)$$

$$T = 3'5 + 0'15 = 3'65\ N$$